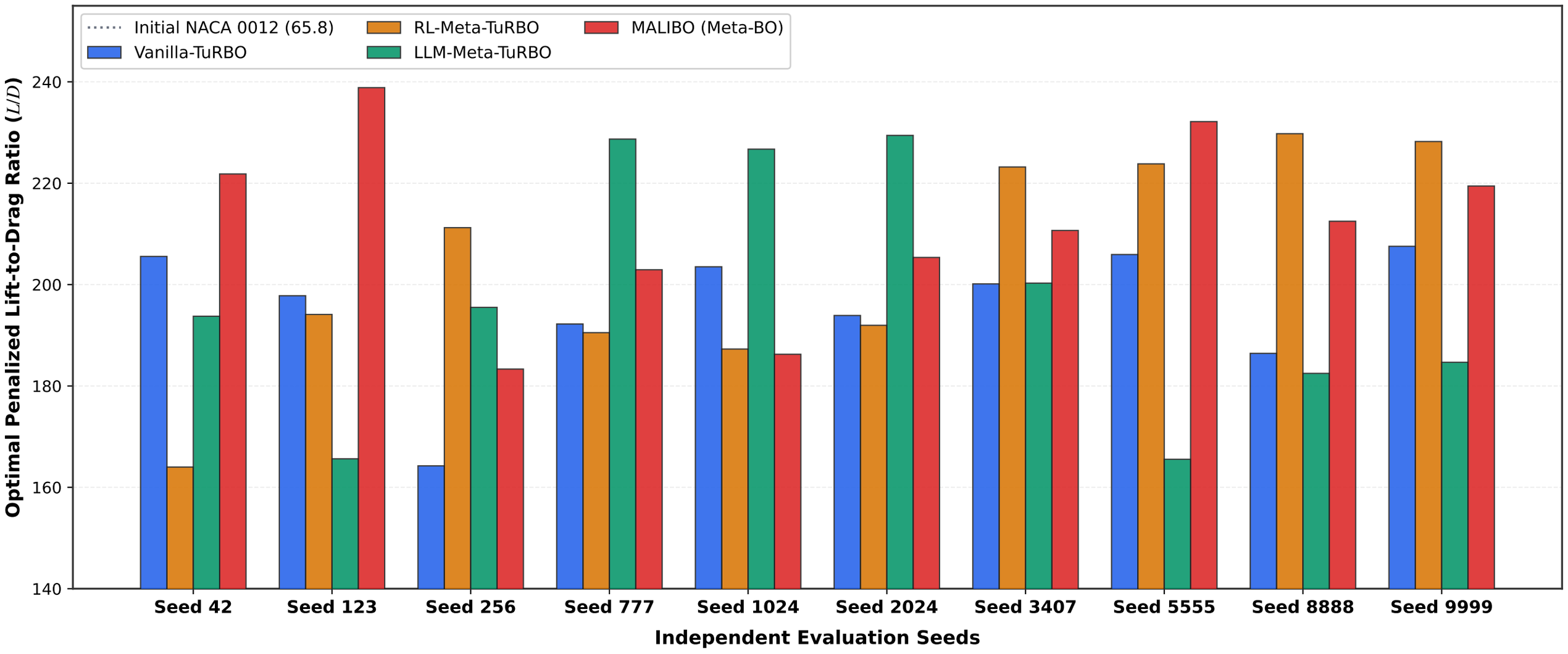
Comparative Benchmark across 10 Independent Seeds
Flow Conditions: $Re = 3.5 \times 10^6, \alpha = 3.5^\circ$ | NFE = 300
Initial NACA 0012 (65.8)
Vanilla-TuRBO
RL-Meta-TuRBO
LLM-Meta-TuRBO
MALIBO (Meta-BO)
Optimal Penalized Lift-to-Drag Ratio (L/D)
240
220
200
180
160
140
Seed 42
Seed 123
Seed 256
Seed 777
Seed 1024
Seed 2024
Seed 3407
Seed 5555
Seed 8888
Seed 9999
Independent Evaluation Seeds